Heer Kumari Perchani

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EDUCATION

University of Alberta | *Bachelor of Science, Major in Computing Science, Minor in Statistics* **Expected May 2027**
Awards: Dean's Honor Roll, First Class Standing (GPA: 3.8/4.0)
Relevant Coursework: Computer Vision, NLP, Machine Learning, Data Structures and Algorithms, Probability and Statistics

WORK EXPERIENCE

Community Caretaker | *St. Joseph's College Edmonton, AB* **Sep 2025 – Present**

- Lead and coordinate community events**, resolve resident concerns,** and maintain a collaborative living environment
- **Communicate** with residents and stakeholders to **address issues efficiently** in a dynamic, fast-paced setting

Software & Data Analyst Intern| *SoftDev, Remote* **June 2025 – Present**

- Developed and tested **SQL Server** queries and parameterized reports supporting advertisement scheduling, billing, and financial modules (GL, vouchers, trial balance).
- Performed **data transformation** and manipulation during new client onboarding to standardize contracts and financial data.
- Currently contributing to building and maintaining the company website to showcase its Windows based ERP solutions.

Teaching Assistant CMPUT 175 (Python) | *University of Alberta, Edmonton, AB* **Sep 2025 – Dec 2025**

- Facilitated technical labs and debugged Python codebases for **130+ students**, ensuring mastery of core Algorithms.
- **Evaluated** complex programming assignments, providing architectural feedback to improve code efficiency and readability

PROJECTS

Netflix Movies & TV Shows Dashboard | Power BI

- Cleaned and engineered **8,278 Netflix titles** in Power Query (handling missing data, imputing countries, feature creation, sentiment tagging) to build a reproducible, analysis-ready dataset.
- Developed an interactive Power BI dashboard (area, bar, tree map, pie, KPI cards) to uncover trends and deliver actionable insights on genres, ratings, and global content distribution.

Human Emotion Classification Web Application

- Engineered a **full-stack deep learning application** using **PyTorch** for the backend and a **Flask-based web interface**
- Developed an end-to-end pipeline to process **facial image datasets** and classify them into 5 distinct emotion categories
- Trained a **Convolutional Neural Network (CNN)** to achieve 71% accuracy, optimizing the model for real-time inference 2during live user interaction

Feelin' | Android Mood-Sharing Application

- Co-developed a **data-driven Android application** utilizing **Java** and **NoSQL** to facilitate real-time emotional tracking and social sharing
- Integrated **Firebase** for secure data retrieval of personal mood histories and synchronized global feeds, while implementing the **Google Maps API** to visualize regional mood trends via localized emojis
- Enhanced user accessibility by designing **interactive XML UI/UX layouts** that streamlined the mood-logging workflow

Chrome Extension | Head Movement Navigation

- Architected a **computer-vision Chrome extension** using JavaScript and FaceAPI.js to **facilitate hands-free web accessibility**
- Engineered custom motion-detection algorithms to **translate head movements into browser commands** such as scrolling and page navigation
- Awarded **DivE Category Winner at HackED 2025** for improving digital inclusivity through high-accuracy calibration and responsive interaction design

CERTIFICATIONS

Google Data Analytics Specialization (Coursera) **July 2024**

- Completed an immersive program covering data cleaning, analysis, and visualization using spreadsheets, SQL, R, and Tableau, developing practical skills used by data analysts in day-to-day work

SKILLS

Languages: Python, SQL, JavaScript, R, C/C++, Kotlin, Java, HTML, CSS, Bash, Regex
Frameworks and Libraries: NumPy, PyTorch, Tensor Flow, OpenCV, FastAPI, Pandas, Matplotlib, Scikit-Learn, Seaborn
Developer Tools: PostgreSQL, SQLServer**,** Firebase, GitHub, VS Code, Docker, Power BI, Tableau, IBM SPSS, Microsoft Excel